

Objectives

- Explain React events
- Explain about event handlers
- Define Synthetic event
- Identify React event naming convention

In this hands-on lab, you will learn how to:

- Implement Event handling concept in React applications
- Use this keyword
- Use synthetic event

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **90 minutes.**

Create a React Application “eventexamplesapp” to handle various events of the form elements in HTML.

1. Create “Increment” button to increase the value of the counter and “Decrement” button to decrease the value of the counter. The “Increase” button should invoke multiple methods.
 - a. To increment the value
 - b. Say Hello followed by a static message.



2. Create a button “Say Welcome” which invokes the function which takes “welcome” as an argument.



3. Create a button which invokes synthetic event “OnPress” which display “I was clicked”



Create a “CurrencyConvertor” component which will convert the Indian Rupees to Euro when the Convert button is clicked.

Handle the Click event of the button to invoke the handleSubmit event and handle the conversion of the euro to rupees.



```
Windows PowerShell
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\11. ReactJS-HOL> npx create-react-app eventexamplesapp

Creating a new React app in C:\Users\anana\Downloads\Digital Nurture\Week 5\React\11. ReactJS-HOL\eventexamplesapp.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...

added 1301 packages in 1m

272 packages are looking for funding
  run `npm fund` for details

Installing template dependencies using npm...

added 17 packages, removed 2 packages, and changed 1 package in 6s

271 packages are looking for funding
  run `npm fund` for details
Removing template package using npm...

removed 1 package, and audited 1316 packages in 7s

271 packages are looking for funding
  run `npm fund` for details

28 vulnerabilities (9 low, 6 moderate, 13 high)
```

APP.JS

```
import React, { useState } from "react";
```

```
function App() {
```

```
  const [count, setCount] = useState(0);
```

```
  const [amount, setAmount] = useState("");
```

```
  const [currency, setCurrency] = useState("");
```

```
  // Increment
```

```
  function incrementValue() {
```

```
    setCount(count + 1);
```

```
  }
```

```
  // Decrement
```

```
  function decrementValue() {
```

```
    setCount(count - 1);
```

```
  }
```

```
// Hello Message
```

```
function sayHello() {  
  alert("Hello! Member!");  
}
```

```
// Welcome Message
```

```
function sayWelcome(message) {  
  alert(message);  
}
```

```
// Synthetic Event
```

```
function handleClick() {  
  alert("I was clicked");  
}
```

```
// Currency Converter
```

```
function handleSubmit(e) {  
  
  e.preventDefault();  
  
  const euro = amount * 80;  
  
  alert("Converting to Euro Amount is " + euro);  
}  
  
return (
```

```
<div style={{ margin: "30px" }}>
```

```
<h2>{count}</h2>
```

```
{/* Increment invokes two methods */}
```

```
<button  
  onClick={() => {  
    incrementValue();  
    sayHello();  
  }}  
>
```

```
  Increment  
</button>
```

```
<br /><br />
```

```
<button onClick={decrementValue}>  
  Decrement  
</button>
```

```
<br /><br />
```

```
<button  
  onClick={() => sayWelcome("welcome")}  
>  
  Say welcome  
</button>
```

```
<br /><br />
```

```
<button
  onClick={handleClick}
>
  Click on me
</button>
```

```
<br /><br />
```

```
<h1 style={{ color: "green" }}>
  Currency Convertor!!!
</h1>
```

```
<form onSubmit={handleSubmit}>
```

```
<label>Amount: </label>
```

```
<input
  type="number"
  value={amount}
  onChange={(e) => setAmount(e.target.value)}
/>
```

```
<br /><br />
```

```
<label>Currency: </label>
```

```
<input
  type="text"
```

```
    value={currency}
    onChange={(e) => setCurrency(e.target.value)}
  />
```

```
<br /><br />
```

```
<button type="submit">
  Submit
</button>
```

```
</form>
```

```
</div>
```

```
);
```

```
}
```

```
export default App;
```




